

# Ryo ISHIZUKA | 石塚 伶

Institute of Science Tokyo

2-12-1 Ookayama 152-8551 Meguro-ku Tokyo – Japan

✉ ishizuka.r.ac@m.titech.ac.jp • <https://ryo1203.github.io>

## Education

|                                                                                                                                                                   |                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <b>Institute of Science Tokyo (formerly Tokyo Institute of Technology)<sup>1</sup></b><br><i>Doctor of Science in Mathematics</i><br>Supervisor: Kazuma Shimomoto | <b>Tokyo, Japan</b><br><i>Apr 2024–Mar 2026</i> |
| <b>Tokyo Institute of Technology</b><br><i>Master of Science in Mathematics</i><br>Supervisor: Kazuma Shimomoto (2nd year), Fumiharu Kato (1st year)              | <b>Tokyo, Japan</b><br><i>Apr 2022–Mar 2024</i> |
| <b>Tokyo Institute of Technology</b><br><i>Bachelor of Science in Mathematics</i><br>Supervisor: Fumiharu Kato                                                    | <b>Tokyo, Japan</b><br><i>Apr 2018–Mar 2022</i> |

## Professional Position

|                                                                                                                |                          |
|----------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>JSPS Research Fellow (PD)</b><br><i>Institute of Science Tokyo</i><br>Host Researcher: Kazuma Shimomoto     | <b>Apr 2026–Mar 2027</b> |
| <b>JSPS Research Fellow (DC1)</b><br><i>Tokyo Institute of Technology</i><br>Host Researcher: Kazuma Shimomoto | <b>Apr 2024–Mar 2026</b> |

## Research Interests

Commutative algebra in mixed characteristic (via arithmetic methods such as perfectoid rings, prismatic cohomology, and almost mathematics).

## Papers and Preprints

11. **R. Ishizuka** and S. Yoshikawa, “A local-global correspondence for perfectoid purity”, arXiv:2604.25265, 2026.
10. **R. Ishizuka** and S. Yoshikawa, “Algebraization of absolute perfectoidization via section rings”, arXiv:2604.02682, 2026.
9. **R. Ishizuka** and S. Yoshikawa, “Derived graded modules”, arXiv:2601.19164, 2026.
8. **R. Ishizuka** and S. Yoshikawa, “Graded perfectoid rings”, arXiv:2511.02322, 2025.
7. **R. Ishizuka** and K. Shimomoto, “Quasi-canonical lifting of projective varieties in positive characteristic”, arXiv:2506.01345, 2025.
6. **R. Ishizuka**, “A higher algebraic approach to liftings of modules over derived quotients”, arXiv:2503.17964, 2025.
5. **R. Ishizuka**, “Perfectoid towers generated from prisms”, Nagoya Math. J. **261**, e17 (2026). arXiv:2409.15785, 2024.
4. **R. Ishizuka** and K. Nakazato, “Prismatic Kunz’s theorem”, J. Algebra **693**, 732–769 (2026). arXiv:2402.06207, 2024.

<sup>1</sup>In Oct 2024, Tokyo Institute of Technology was merged with Tokyo Medical and Dental University and reorganized into *Institute of Science Tokyo*.

3. D. Dine and **R. Ishizuka**, “*Tilting and untilting for ideals in perfectoid rings*”, Math. Z. **307**, 66 (2024). arXiv:2308.09600, 2023.
2. **R. Ishizuka**, “*A calculation of the perfectoidization of semiperfectoid rings*”, Nagoya Math. J. **255**, 742-759 (2024). arXiv:2305.07916, 2023.
1. K. Shimomoto and **R. Ishizuka**, “*A mixed characteristic analogue of the perfection of rings and its almost Cohen-Macaulay property*”, arXiv:2303.13872, 2023.

## Talks

---

36. Dec 2025. “*Global singularities in mixed characteristic and graded perfectoid rings*”, Mini-workshop on Arithmetic Geometry in Sendai, Tohoku University, Japan.
35. Nov 2025. “*Perfectoid towers arising from Frobenius lifts*”, The 46th Japan Symposium on Commutative Algebra, Hotel Fukuracia Osaka Bay, Japan.
34. Nov 2025. “*Perfectoid towers arising from prisms*”, Workshop on Number theory at Tsuda University 2025, Tsuda University, Japan.
33. Oct 2025. “*Absolute perfectoidization of schemes and local to global principle in mixed characteristic*”, Singularity Theory Seminar, Nihon University, Japan.
32. Sep 2025. “*The Frobenius-Witt cotangent bundle*” (Survey talk), Yatsugatake Workshop, 2025, Wellness Garden in Goddess Forest : Kobuchizawa, Japan.
31. Sep 2025. “*A version of Kunz’s theorem in mixed characteristic*”, Noda Algebraic Geometry Symposium, Tokyo University of Science, Japan.
30. July 2025. “*On the derived deformation functor of Frobenius liftings*” (Poster Session), 2025 Summer Research Institute in Algebraic Geometry., Colorado State University, USA.
29. July 2025. “*On computation of Tor modules over a derived quotient*”, The 36th Seminar on Commutative Algebra in Japan., Kyushu University, Japan.
28. June 2025. “*A unified construction of perfectoid towers by prisms*”,  $p$ -adic and Characteristic  $p$  Methods in Algebraic Geometry., EPFL, Switzerland.
27. May 2025. “*Approximation of perfectoid rings by Noetherian rings and prisms*”, MPIM Algebra Seminar., MPIM, Germany.
26. Mar 2025. “*On the vanishing of Ext and liftings of modules on derived quotients using higher algebras*”, MSJ Spring Meeting 2025., Waseda University, Japan.
25. Mar 2025. “*Frobenius liftability and derived algebraic geometry*”, The 29th Conference on Algebra for Young Researchers in Japan., Osaka University, Japan.
24. Mar 2025. “*Liftability results from formal moduli problems*” (including survey), Small workshop on arithmetic geometry 2025 in Hakodate., Hakodate Community Design Center, Japan.
23. Nov 2024. “*Prismatic Kunz’s theorem*”, The 45th Japan Symposium on Commutative Algebra., RIMS (Kyoto University), Japan.
22. Nov 2024. “*Applications of higher algebra to the lifting problem of modules*”, Tokyo Commutative Algebra Seminar., Online (Zoom), Japan.
21. Oct 2024. “ *$F$ -liftability obstruction in singular varieties through derived algebraic geometry*”, Singularity Theory Seminar., Nihon University, Japan.
20. Sep 2024. “*Frobenius maps on mixed characteristic rings via prismatic cohomology*” (Poster Session), L-functions and Motives in Niseko 2024., Setsu Niseko and Niseko Residents Center, Japan.
19. Sep 2024. “*A generalization of Kunz’s theorem to mixed characteristic via  $p$ -adic cohomology theory*”, MSJ Autumn Meeting 2024., Osaka University, Japan.

18. July 2024. *"Perfectoid spaces, tilts and untilts"* (Survey talk), Atelier de Géométrie Arithmétique 2024., RIMS (Kyoto University), Japan.
17. July 2024. *"Prismatic approach to a mixed characteristic Kunz's theorem"*, The 23nd Hiroshima-Sendai Workshop on Number Theory at Sendai., Tohoku University, Japan.
16. June 2024. *"Regularity criterion of mixed characteristic rings via prismatic cohomology"*, Keio Algebra Seminar., Keio University, Japan.
15. June 2024. *"Prisms and regular local rings"*, The 35th Seminar on Commutative Algebra in Japan., Tokushima University, Japan.
14. Apr 2024. *"Prisms and its application to regular rings"*, Saturday Seminar., Meiji University, Japan.
13. Mar 2024. *"Perfectoid ideals and its correspondence"*, The 20th Mathematics Conference for Young Researchers., Hokkaido University, Japan.
12. Feb 2024. *"Mixed characteristic Kunz's theorem with prismatic theory"*, The 28th Conference on Algebra for Young Researchers in Japan., Waseda University, Japan.
11. Dec 2023. *"Commutative ring theoretic approach for the perfectoidization of semiperfectoid rings"*, Number Theory Seminar., Kyoto University, Japan.
10. Nov 2023. *"Ideal correspondence between a perfectoid ring and its tilt"*, The 44th Japan Symposium on Commutative Algebra., Lectore Hayama, Japan.
9. Aug 2023. *"Absolute integral closure"* (Survey talk), The 18th Summer School on Commutative algebra., Tokyo Institute of Technology, Japan.
8. Aug 2023. *"On the relation between perfectoidization and  $p$ -root closure"*, The 9th China-Japan-Korea International Conference on Ring and Module Theory., Incheon National University, Republic of Korea.
7. July 2023. *"On the commutative ring-theoretic structure of the perfectoidization of semiperfectoid rings"*, The 22nd Hiroshima-Sendai Workshop on Number Theory at Hiroshima., Hiroshima University, Japan.
6. July 2023. *"On the application of perfectoidization to commutative algebra and its structure"*, The 34th Seminar on Commutative Algebra in Japan., Kitami Institute of Technology, Japan.
5. May 2023. *"On Perfectoid(ization) and its commutative ring-theoretic properties"*, Ookayama Youth Seminar in Algebra., Tokyo Institute of Technology, Japan.
4. Mar 2023. *"A mixed characteristic analogue of the perfection of rings"*, The 11th Japan-Vietnam Joint Seminar on Commutative Algebra - by and for young mathematicians -, Vietnam Academy of Science and Technology, Vietnam.
3. Mar 2023. *"An explicit construction of perfectoid almost Cohen-Macaulay algebras in mixed characteristic"*, MSJ Spring Meeting 2023., Chuo University, Japan.
2. Mar 2023. *"An explicit construction of perfectoid almost Cohen-Macaulay algebras"*, The 19th Mathematics Conference for Young Researchers., Hokkaido University, Japan.
1. Nov 2022. *"An explicit construction of perfectoid almost Cohen-Macaulay algebras in mixed characteristic"*, The 43rd Japan Symposium on Commutative Algebra., Osaka University, Japan.

## Membership

---

- Apr 2023– . Mathematical Society in Japan (MSJ)

## Languages

---

**Japanese:** Native

*Mother tongue*

**English:** Intermediate

*Can read, write, and, listen but may struggle with conversation*

**French:** Beginner

*Can only read mathematical texts*